Density-Matrix Digital Twin for Noisy Quantum Circuits

A simulation study of relaxation, dephasing, gate errors, readout effects, and uncertainty propagation

Abstract

A three-qubit density-matrix simulator was implemented to represent ideal gate operations and selected sources of noise in quantum circuits. Single-qubit gates were embedded into an eight-dimensional computational space, while time-dependent amplitude damping and phase-noise channels were applied through Kraus operators. The workflow included state checks, a readout-confusion model, synthetic T_1 inversion-recovery, Ramsey-style and Hahn-echo sequences, pulse-amplitude error fitting, residual inspection, a readout/SPAM-augmented T_1 fit, simple circuit execution, and Monte Carlo propagation of fitted-parameter uncertainty into output probabilities. Curve fitting returned T_1 = 50.00 μs, T_2* = 50.00 μs, and T_2,echo = 50.00 μs for the synthetic traces; the pulse-train fit estimated an angle error of 1.005° ± 0.00047°. These values are consistent with the parameters used to generate the synthetic data, but they should not be interpreted as experimental device characterization. Several validation and sensitivity-analysis sections use placeholder values, and the T_1 validation compares the simulated signal with an incorrect analytic target. The report therefore distinguishes numerical outputs from claims that require corrected validation or real-device measurements.

1. Introduction

Quantum processors are affected by energy relaxation, loss of phase coherence, imperfect control pulses, and measurement errors. A density-matrix representation is suitable for modeling mixed states and non-unitary noise processes. In this study, a compact three-qubit simulator was developed to explore how these mechanisms can be represented and how estimated noise parameters can be carried into predictions for circuit output distributions. The scope is a classical numerical prototype rather than a hardware experiment.

2. Methods

2.1 State representation and gates

The initial state was $|000\rangle$, represented by an 8 × 8 density matrix ρ with $\rho_{00} = 1$. The identity (I), Pauli-X (X), Pauli-Z (Z), and Hadamard (H) matrices were defined explicitly. A one-qubit gate U acting on qubit q was embedded into the three-qubit Hilbert space using Kronecker products, then applied as $\rho' = U\rho U\dagger$. The simulator calculated the expectation value of a local Z observable from $\langle Zq\rangle = \mathrm{Tr}(Zq\rho)$.

2.2 Noise channels

Amplitude damping was parameterized by $p_1(\tau) = 1 - \exp(-\tau/T_1)$, with Kraus operators $K_0 = \mathrm{diag}(1, \sqrt{(1 - p_1)})$ and $K_1 = [[0, \sqrt{p_1}], [0, 0]]$. A phase-flip/dephasing channel was applied using $E_0 = \sqrt{(1 - p\varphi)}I$ and $E_1 = \sqrt{p\varphi}\, Z$, where $p\varphi = \frac{1}{2}[1 - \exp(-\tau/T\varphi)]$. Each channel was applied locally to a selected qubit, with the other qubits acted on by identity operators.

The intended relaxation/coherence relationship was $1/T_2 = 1/(2T_1) + 1/T\varphi$. For the parameters T_1 = 50 and $T\varphi$ = 100 time units, this gives T_2 = 50 time units. The notebook labels the fitted times in microseconds; this unit is meaningful only if the simulated time axis is calibrated to microseconds.

2.3 Measurement and readout model

Ideal computational-basis probabilities were taken from the diagonal of ρ. A seeded, near-diagonal 8 × 8 confusion matrix was normalized column-wise and multiplied by the ideal probability vector to produce readout-distorted probabilities.

2.4 Calibration experiments and fitting

Synthetic traces were generated for T_1 inversion recovery (30 delays from 0 to 200), a Ramsey-style sequence (40 delays from 0 to 150), a Hahn-echo sequence (40 delays from 0 to 150), and a pulse train with a 1° rotation-angle error. Nonlinear least-squares fitting used exponential models for T_1 and echo decay, a damped cosine for the Ramsey-style trace, and a cosine-based model for pulse-train error. Approximate 95% fit intervals were calculated as 1.96 times the standard error derived from the fitted covariance matrix.

2.5 Circuit prediction and uncertainty propagation

Two illustrative circuits were encoded as operation lists. The notebook sampled fitted T_1, Ramsey decay, and pulse-error parameters 200 times from a multivariate normal distribution using a diagonal covariance approximation. For each sample, the circuit was simulated and the eight basis-state probabilities were recorded. The median and 5th/95th percentiles were used as the prediction and nominal 90% interval. A total-variation-distance (TVD) function was also defined for comparing predicted and reference probability vectors.

3. Results

3.1 Basic implementation checks

The initial density matrix had shape 8 × 8 and unit population in |000⟩. Applying X to qubit 0 placed the population at matrix index [4, 4], consistent with the notebook's qubit ordering and the state label |100⟩. The readout-confusion test retained the maximum probability at index 4, with a reported peak of 0.9805.

The printed T_1 validation gave a simulated $\langle Z \rangle = 0.2642$ at $\tau = T_1 = 50$, whereas the notebook's analytic target was $-\exp(-1) = -0.3679$. For an initially excited state relaxing toward |0⟩, the expected Z signal is $1 - 2\exp(-\tau/T_1)$, which equals approximately 0.2642 at $\tau = T_1$. Thus the simulation agrees with the standard relaxation expression, while the validation target in the notebook is incorrect and the test should be revised.

The trace/positivity check printed a pass message. However, the validation function computes the trace but does not actually assert that $\mathrm{Tr}(\rho) = 1$; it only raises an error if an eigenvalue is sufficiently negative. The trace-preservation claim therefore requires an explicit numerical assertion.

3.2 Fitted calibration parameters

Parameter	Fitted value	Interpretation
T_1	$50.00 \pm 4.43 \times 10^{-15}$ μs	Synthetic inversion-recovery fit
T_2^*	$50.00 \pm 1.11 \times 10^{-8}$ μs	Synthetic Ramsey-style fit
$T_{2,\mathrm{echo}}$	$50.00 \pm 7.62 \times 10^{-15}$ μs	Synthetic echo-model fit
Pulse-angle error	1.005° ± 0.00047°	Pulse-train fit

The fitted values recover the synthetic parameter choices closely. The extremely small uncertainties are a consequence of deterministic, noise-free simulated traces and should not be read as realistic experimental confidence intervals. The Ramsey and echo sequences in the current implementation also omit several experimental features (for example, sampled measurement noise and a realistic detuning/noise ensemble), so their labels should be understood as idealized demonstrations.

3.3 SPAM-augmented fit and model comparison

The T_1 fit extended with visibility and offset returned T_1 = 50.0000, visibility = 1.0000, and offset ≈ 0.0000. The fitting routine emitted an OptimizeWarning that parameter covariance could not be estimated. The AIC values were −2198.02 for the one-parameter model and −1564.09 for the three-parameter SPAM model. The lower AIC favors the simpler model for these synthetic data; the extra parameters did not improve the fit enough to offset the complexity penalty.

3.4 Circuit-level examples

A GHZ-related state-preparation sequence produced a reported Z expectation of 0.0 for qubit 0, consistent with a balanced superposition for that qubit. A randomized low-depth versus high-depth example printed ⟨Z⟩ = −0.6375 and −0.4165, respectively. Since the circuit uses Python's random module and no fixed seed is shown, these values are one illustrative run rather than reproducible benchmarks.

3.5 Uncertainty intervals and TVD sensitivity

The notebook generated nominal 90% intervals for basis-state probabilities from 200 Monte Carlo samples and plotted intervals for the first mock circuit. The reference probability vector used in the TVD sensitivity section is explicitly a placeholder concentrated entirely in |000⟩, not a measured or independently validated target distribution. The T_1 sensitivity derivative was reported as 2.8314×10^{-7} per parameter unit, and its placeholder-based contribution was 2.8314×10^{-6}. Consequently, the parameter-contribution bar chart is a pipeline demonstration, not a reliable attribution of error against hardware truth.

4. Figures

The following figures are the plot outputs stored in the supplied notebook. Captions describe the intended analysis; interpretation is limited where the notebook uses synthetic data or placeholder reference values.

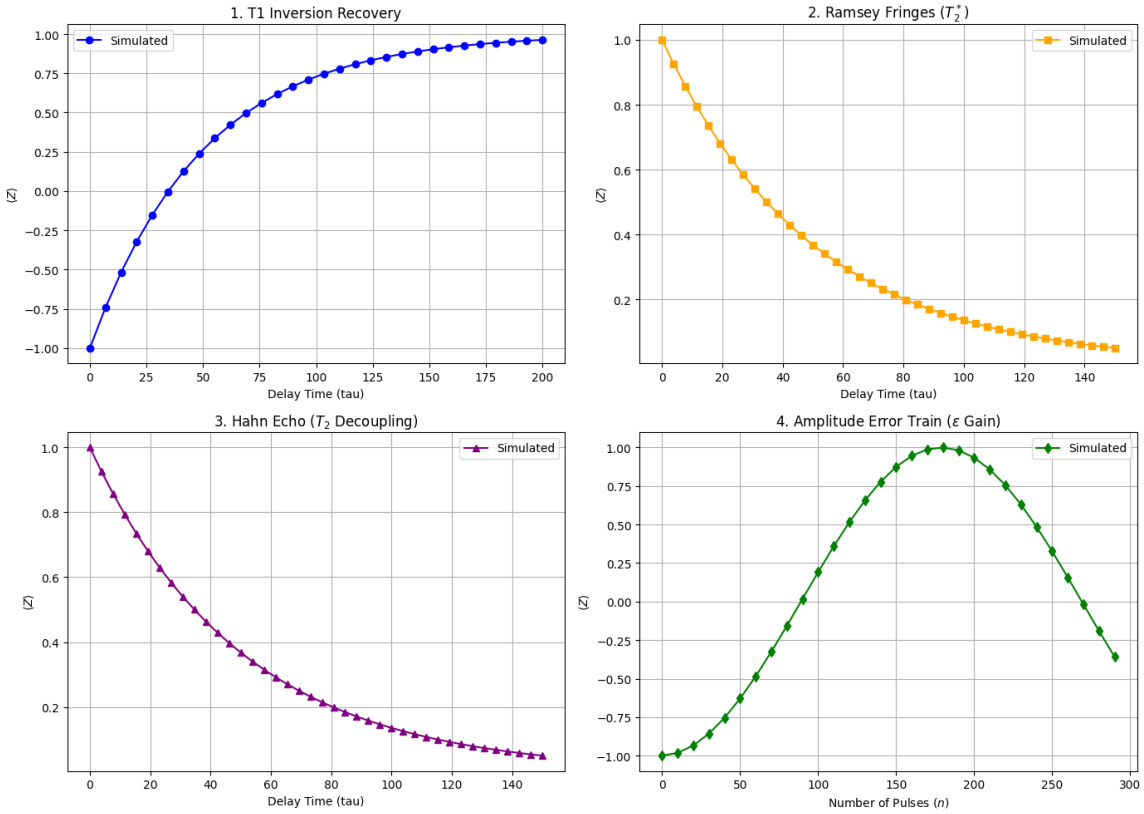


Figure 1. Calibration traces: T_1 inversion recovery, Ramsey fringes, Hahn echo, and pulse-amplitude error amplification.

What it signifies: The four panels summarize how the simulated qubit signal changes with delay time or repeated imperfect pulses. The T_1 panel tracks energy relaxation from the excited state toward equilibrium. The Ramsey panel shows oscillatory coherence loss, while the Hahn-echo panel shows coherence after a refocusing pulse. The pulse-train panel illustrates how a small rotation-angle error can accumulate as the number of pulses increases. These are synthetic simulations, so they demonstrate the behavior of the model rather than characterize a physical device.

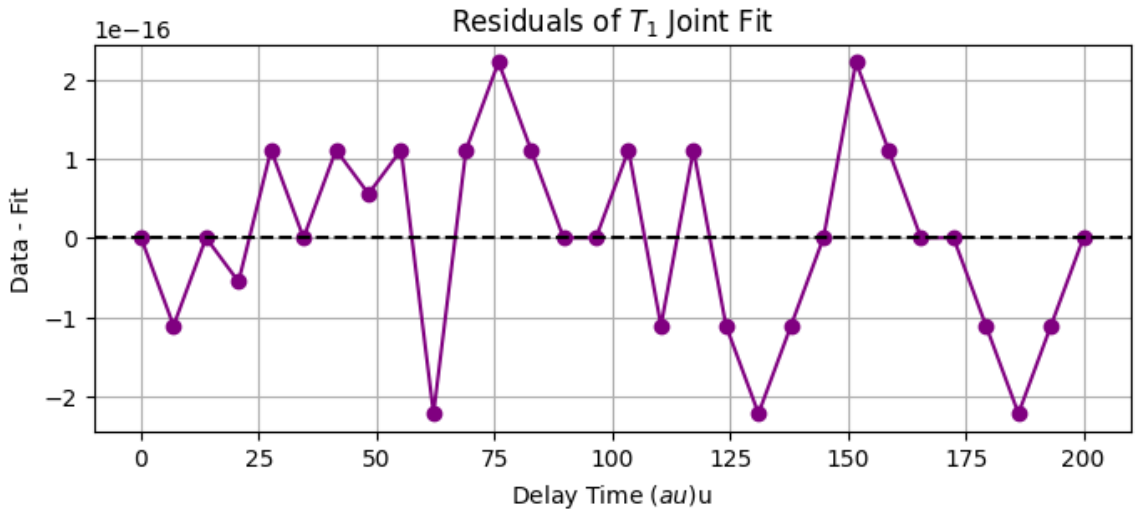


Figure 2(a). Residuals of the T_1 relaxation fit.

What it signifies: Each point is the simulated signal minus the fitted T_1 curve. Values close to the horizontal zero line indicate that the model reproduces the simulated data well. A systematic curve or trend would suggest that the exponential model misses some behavior; scattered deviations around zero would be more consistent with random measurement noise. Here, the trace is synthetic and essentially noise-free, so very small residuals are expected.

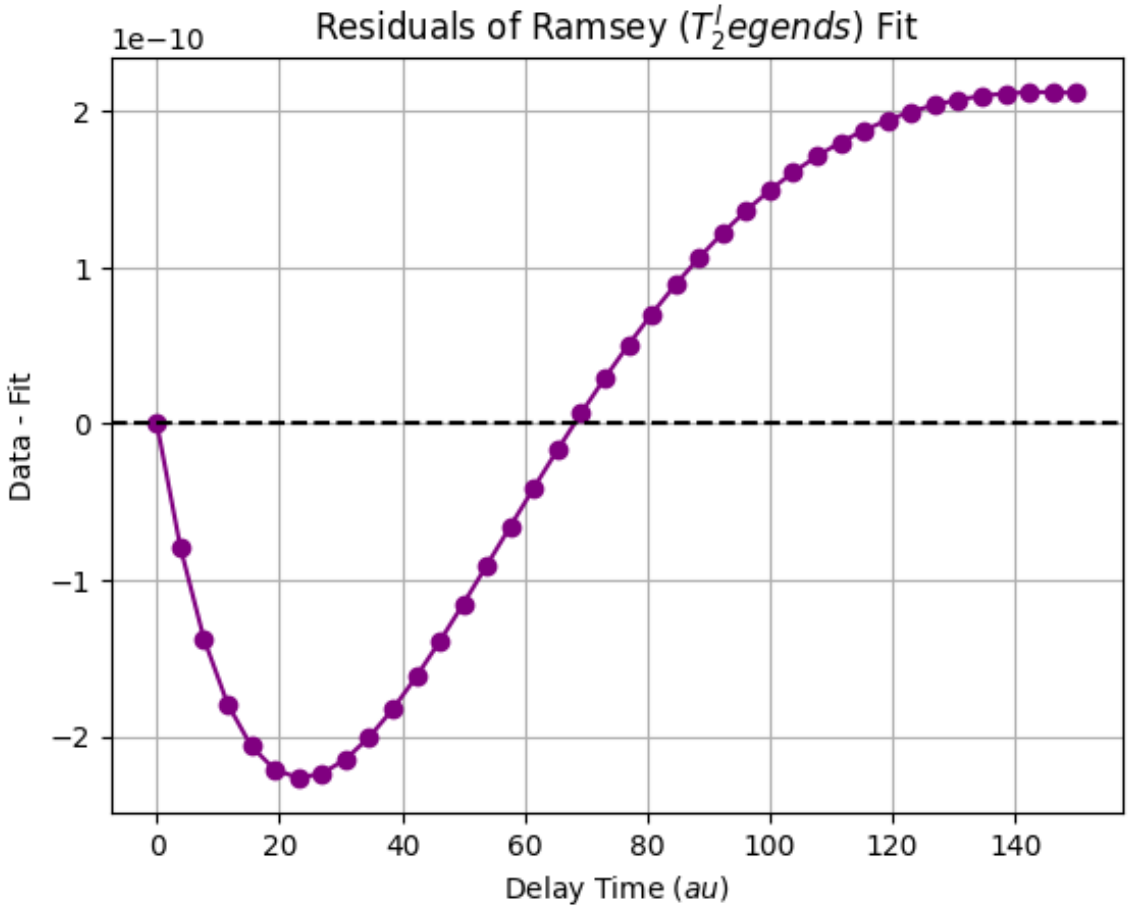


Figure 2(b). Residuals of the Ramsey-style T_2* fit.

What it signifies: The residuals show whether the damped-oscillation model captures both the Ramsey oscillation and its decaying envelope. Residuals near zero indicate agreement between the model and simulated trace. Repeated oscillatory structure in the residuals would suggest a mismatch in frequency, phase, or decay rate. Since the input trace is synthetic, the residuals mainly test numerical consistency.

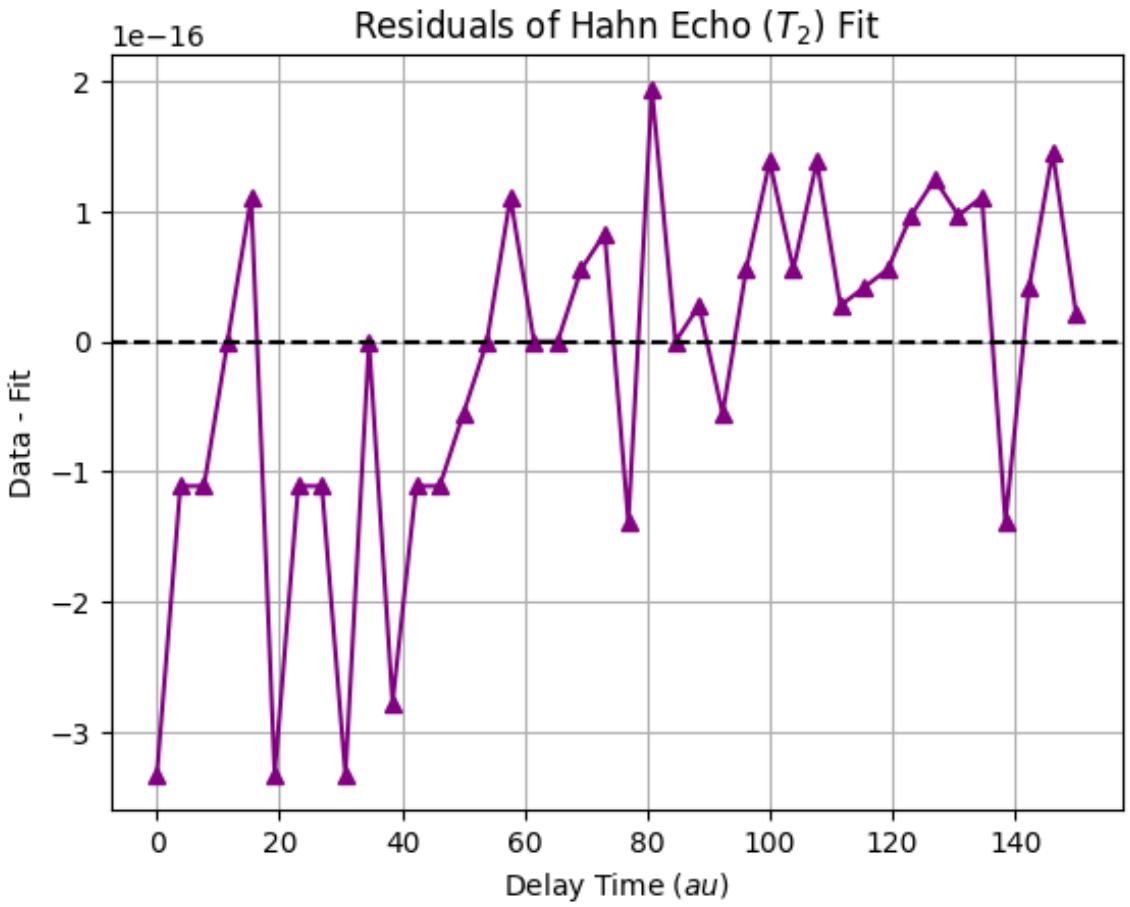


Figure 2(c). Residuals of the Hahn-echo fit.

What it signifies: This plot measures the difference between the simulated echo signal and its fitted decay model. A flat residual pattern around zero supports the selected functional form for the generated data. A persistent positive or negative offset, or a time-dependent trend, would indicate a model mismatch. It does not by itself prove that the same model accurately describes experimental noise.

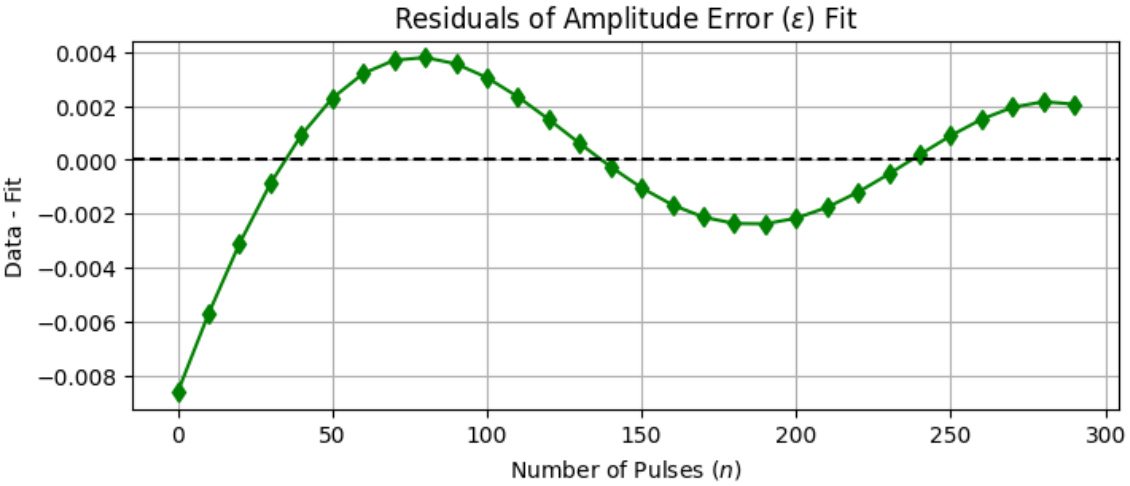


Figure 2(d). Residuals of the pulse-amplitude-error fit.

What it signifies: These residuals compare the pulse-train simulation with the curve used to estimate the rotation-angle error. Small residuals indicate that the fit captures the trend in accumulated pulse error.

Structured residuals would suggest that the pulse-error model or sweep range needs revision. The fit is based on an idealized, deterministic simulation.

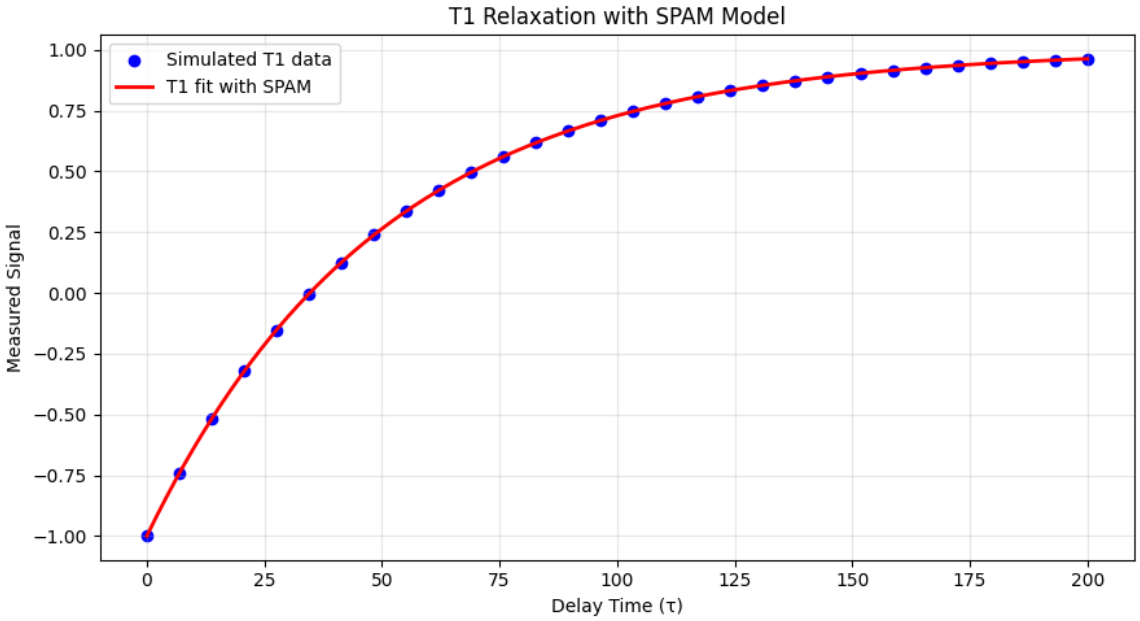


Figure 3. T_1 relaxation fitted with a SPAM-augmented model.

What it signifies: The points show the simulated relaxation signal, while the curve includes visibility and offset parameters intended to represent state-preparation-and-measurement (SPAM) imperfections. The fitted values were approximately $T_1 = 50$, visibility = 1, and offset = 0, meaning that the synthetic trace did not require a measurable visibility reduction or offset. The fit also produced a covariance warning, so the apparent precision should not be interpreted as a realistic experimental uncertainty.

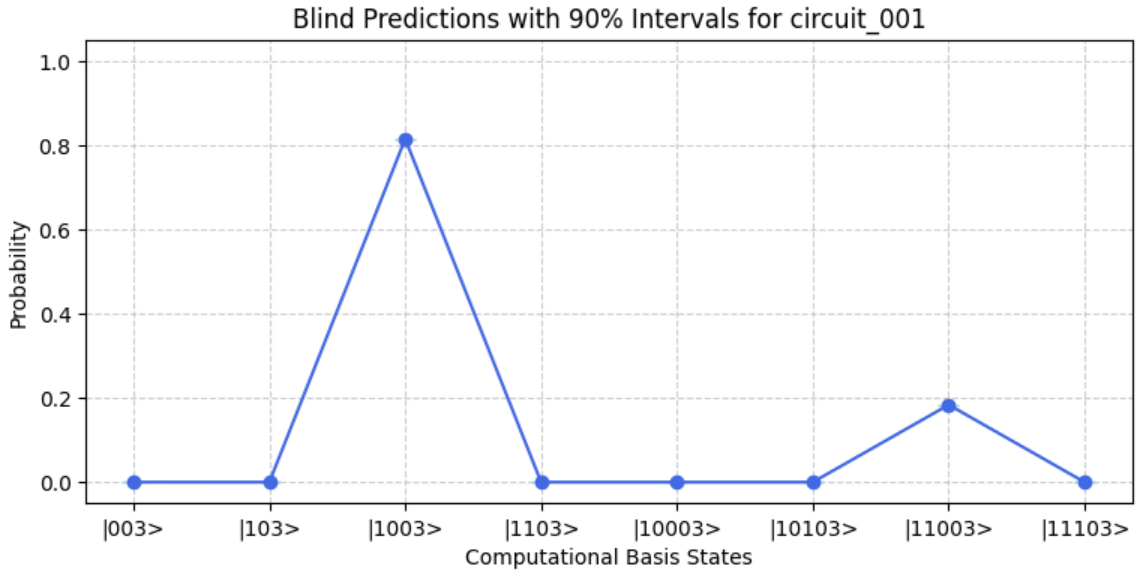


Figure 4. Predicted computational-basis probabilities with nominal 90% Monte Carlo intervals.

What it signifies: Each marker is the median predicted probability of a three-qubit basis state, and the vertical interval spans the 5th to 95th percentiles across parameter samples. Narrow intervals indicate that the sampled parameter uncertainty has little effect on that state's predicted probability; wider intervals indicate greater sensitivity. The intervals reflect the assumptions and sampling procedure used in the simulator, not a statistical confidence interval from hardware measurements.

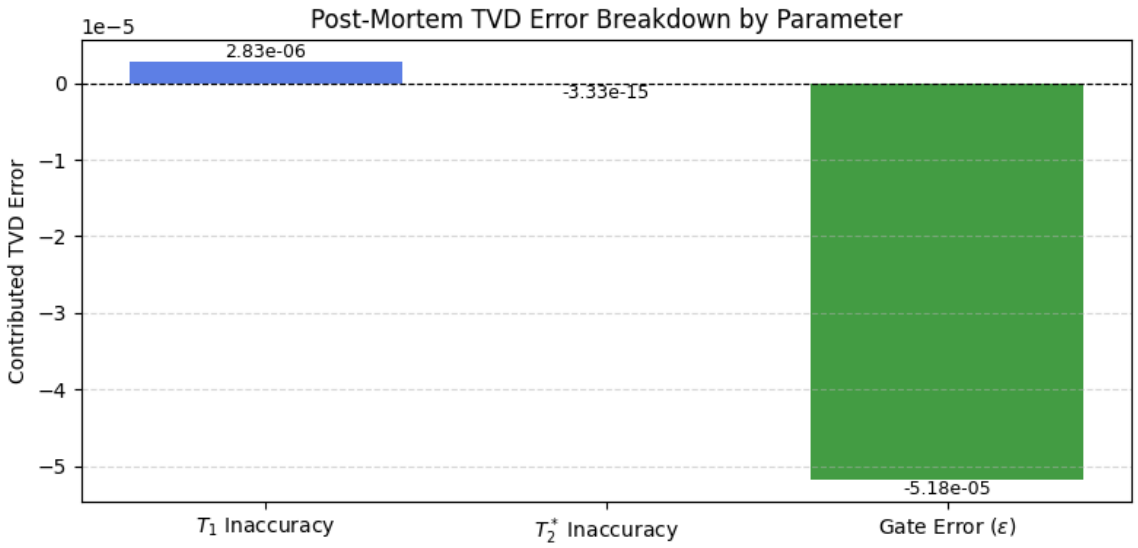


Figure 5. Parameter-wise contribution to the estimated total-variation-distance (TVD) error.

What it signifies: The bars show the estimated change in TVD attributed to uncertainty in T_1, T_2*, and pulse-angle error under the notebook's sensitivity procedure. Larger magnitudes would indicate a parameter with greater estimated influence on the probability-distribution mismatch. However, the reference distribution is a placeholder rather than an independently measured target, so the chart is only a demonstration of the sensitivity-analysis workflow and cannot establish which error source dominates a real device.

5. Discussion and limitations

- Synthetic rather than experimental data: the calibration traces are generated by the same simulator family that is subsequently fitted. The recovered values primarily demonstrate internal consistency.
- Incorrect analytic target in the T_1 validation: replace $-\exp(-\tau/T_1)$ with $1 - 2\exp(-\tau/T_1)$ for an initially excited state under the stated convention $\langle Z\rangle(|0\rangle)=+1$ and $\langle Z\rangle(|1\rangle)=-1$.
- State validation: add explicit checks for trace preservation, Hermiticity, and positive semidefiniteness after every channel.
- Noise-channel parameterization: verify that the phase channel corresponds to the intended $T\varphi$ convention and that combined coherence follows the intended T_2 relation.
- Pulse-train sweep: the comment says 0, 2, 4, …, 18 repetitions, but range(0, 300, 10) actually sweeps 0 through 290 in steps of 10. Correct the comment or the range.
- Qubit mapping: the custom CNOT implementation handles only selected control/target combinations; other combinations fall back to identity. Generalize and test all supported control-target pairs before describing it as a general CNOT simulator.
- Monte Carlo sampling: the covariance is approximated as diagonal, and sampled parameters are not constrained to physically valid ranges. Consider correlated covariance, bounds/truncated distributions, and convergence checks.
- TVD evaluation: replace dummy reference probabilities and placeholder true parameters with independently generated or hardware-measured reference values before drawing conclusions about predictive accuracy.

6. Conclusion

A compact density-matrix simulator was assembled to combine unitary gate operations with Kraus-based relaxation and dephasing, model readout confusion, fit synthetic calibration sequences, and propagate parameter uncertainty into circuit-output probabilities. The fitted synthetic parameters recovered the programmed values, and the workflow produced residual, calibration, prediction-interval, and TVD-contribution plots. The present results establish a useful prototype and analysis pipeline, but they do not yet establish accuracy against a real quantum processor. The most important next steps are to correct the T_1 analytic validation, strengthen physical-state assertions, generalize the CNOT mapping, remove placeholder truth values, and validate the model against independently generated or hardware-derived distributions.